

Pharmacy Desert Risk Prediction Using Machine Learning

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Abstract

The US has experienced a recent wave of closures of pharmacy stores across the country. These closures have created underserved areas, which in turn present an opportunity for either setting-up of new pharmacy businesses or expansion by existing pharmacy businesses. Modeling through High-Order Network models is needed to determining where the pharmacy deserts are likely to occur and how long they are in place prior to exploitation. This would not only offer pharmacy businesses insight on the "where" but also on "when" and how soon to exploit the opportunity presented through pharmacy deserts. The results show that Tensor Train Higher Order Network model had the highest accuracy, at 84% for classifying the exposure risk to pharmacy desert formation with the tot days supply, tot drug cost, tot 30day fills being the top three most important features.

I. INTRODUCTION/BACKGROUND

Successive economic shocks have exposed businesses to disruptions that negatively impact the operations, finance and market confidence. The US has in recent time experienced a wave of closures of pharmacy stores across the country due to the exposures to economic shocks [1], [2]. Closures have resulted in underserved areas forming pharmacy deserts across the United States [3], [4]. However, the pharmacy deserts present an opportunity for either setting-up of new pharmacy businesses or expansion by existing pharmacy businesses [5]. Setting up of pharmacies and businesses would serve to cater for the underserved areas.

II. PROBLEM STATEMENT

Although the closure of businesses creates a gap in the market that can be exploited, questions arise on "when" and "where" the closures occur to create opportunity for entry and expansion. Modeling is needed to determining where the pharmacy deserts are likely to occur and how long they are in place prior to exploitation. This would not only offer pharmacy businesses insight on the "where" but also on "when" and how soon to exploit the opportunity.

III. RELATED WORKS

Disruptions can either be permanent or temporary in terms of impact on the operations, finance and market confidence. Closures and reopening hence characterize disruptions with the modeling of closures requiring factoring of reopening [6]. Temporal approaches form the foundation for modeling of the appearing and disappearing of businesses following disruptions [6]. Economic motivation serves as an important determinant in the opening of businesses and exploitation of business opportunity [7]. XGBoost and Random Forest, provide machine learning approaches that enable the modeling of opening of businesses to exploit business opportunities with considerations to the economic motivations [7]. In addition to the temporal and economic factors, spatial factors are as well important factors in the modeling of the opening and closures of businesses [8]. Likelihood of closure represents a direct and quantitative decision-making indicator that when utilized with machine learning concepts, such as the Support Vector

Machine and Random Forest, provides sector level decision-making insights [9]. Businesses however do not exist in isolation and hence network modeling would improve of the spatial-temporal and economic analysis of the closure of businesses that create deserts, in this case, the health care access network for pharmacy deserts.

IV. METHODOLOGY/APPROACH

Temporal, Spatial and Higher-Order Network Models were be employed to the data for the modeling of risk of pharmacy desert formation across counties the US. Temporal models were observed to be effective in the modeling of appearing and disappearing of business [6]. Consequently, the following Temporal models were employed as baseline models; XGBoost and Long Short-Term Memory (LSTM). Improvement on the temporal features in the baseline included the Spatial-Temporal Graph Neural Network (ST-GNN) and two Higher Order Network models; Hypergraph Neural Network (HGNN) and Tensor Train Network (TT).

V. DATA ANALYSIS

A. Data Acquisition and Data Collection

The acquisition of data involved the collection of data from the following sources:

- Pharmacy Datasets was collected using the CMS API available from the following API link: <https://data.cms.gov/provider-summary-by-type-of-service/medicare-part-d-prescribers/medicare-part-d-prescribers-by-data/2013>. Data was collected for individual years from 2013 to 2023. The resulting data contained 8,294,940 observations. Merging was necessary to combine the data collected for each of the years into the final full pharmacies data.
- Demand-Related Datasets for Health Outcomes and Health Risks was collected from County Health Ranking Data from the following link: <https://www.countyhealthrankings.org/health-data/methodology-and-source-data-documentation/national-data-documentation-2010-2023>. Data was collected for individual years from 2010 to 2023 with the resulting data having a size of 44,216 observations. Merging was as well required to combine the data collected for each of the years into the final full health outcomes and health factors data.
- Economy-Related Datasets for Labor and Income Data was collected from USDA Economic Research Service (ERS) using the following link: <https://www.ers.usda.gov/data-products/county-level-data-sets/county-level-data-sets-download-data>. Data was collected from 2000 to 2023, resulting in a data whose size was 329,725 observations. Although the data was combined for all years, it was in long format and required separation of the labor and income attributes in the data from the attribute column to individual columns. The attributes within the attribute column also had suffix "20xx" for each year; hence, for each attribute there were as many versions as the number of years. Thus, the data cleaning also created a long format data with years in single column.

B. Data Preparation

- Data – Aggregation for CMS Data: The CMS data contained large amount of data and required saving to .parquet format and only accessible in small amounts through Duck Database (Duck DB). Aggregation was conducted within Duck DB so that the aggregates for the numeric variables were created using year, state and county. In creating the target variable, pharmacy per county year, pharmacy location (using street 1 and street 2 variables) was used as identifier for counting number of pharmacies in the year, state and county grouping in the database.
- Data – Data Cleaning and Data Merging: Following the aggregation of the CMS data from the Duck database, the aggregated data was merged with the Health Outcomes and Health Factors dataset, and USDA labor data using year, state and county. Data cleaning involved imputation for missing entries using Q1 within year, state and county groupings.

The summary of the Data aggregation for CMS Data and Data Cleaning and Data Merging is presented below in Figure 1.

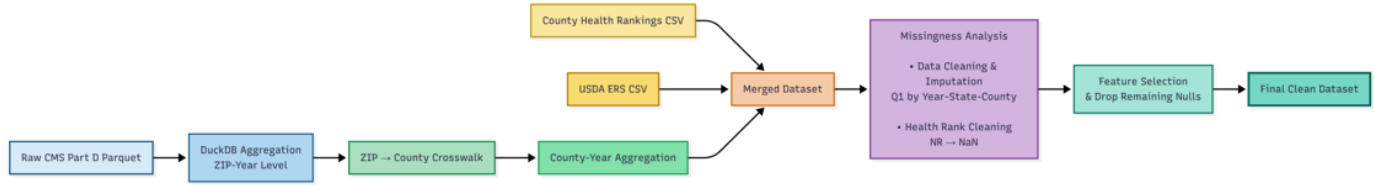


Fig. 1.

- **Feature Engineering:** Data aggregation for the CMS data in Duck DB formed the first stage of feature engineering, the second stage involved the creation of the risk of pharmacy desert formation target variable; Desert Formation Risk. Figure 2 below summarizes the formation of the Desert Formation Risk target variable: as risk of a county being a pharmacy desert across successive years; High Risk and Low Risk. The TDF Feature is such that: Negative values = accessibility improved, Near zero = stable accessibility, Positive values = worsening accessibility, Large positive values = rapid deterioration (potential desert formation). Pharmacy utilization shift of up to 25% provides a threshold for a change in population access of pharmacies [10]. Hence, an increase of 25% or more in the customer load per pharmacy between successive years exposes the county high risk of being a pharmacy desert; forming the logic for the Desert Formation Risk levels.

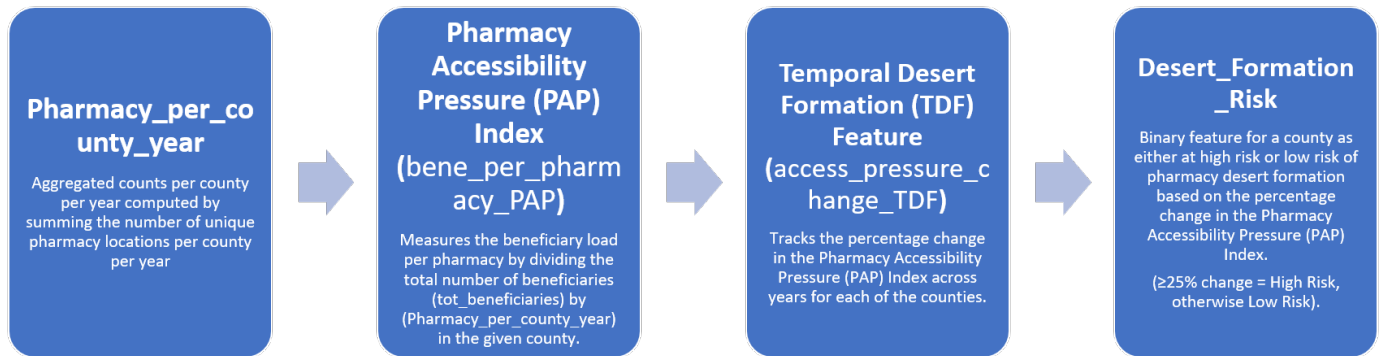


Fig. 2.

C. Exploratory Data Analysis (EDA)

The evolution of the average pharmacy access pressure per state from 2014 to 2023 shown in the bar chart on the left and the slope chart on the right. The states of New Jersey, Kansas and Alaska had the top 3 highest average increases in the pharmacy access pressure. Alaska and New Jersey also show consistency in being the top two states with highest average pharmacy access pressure in 2014 and in 2023.

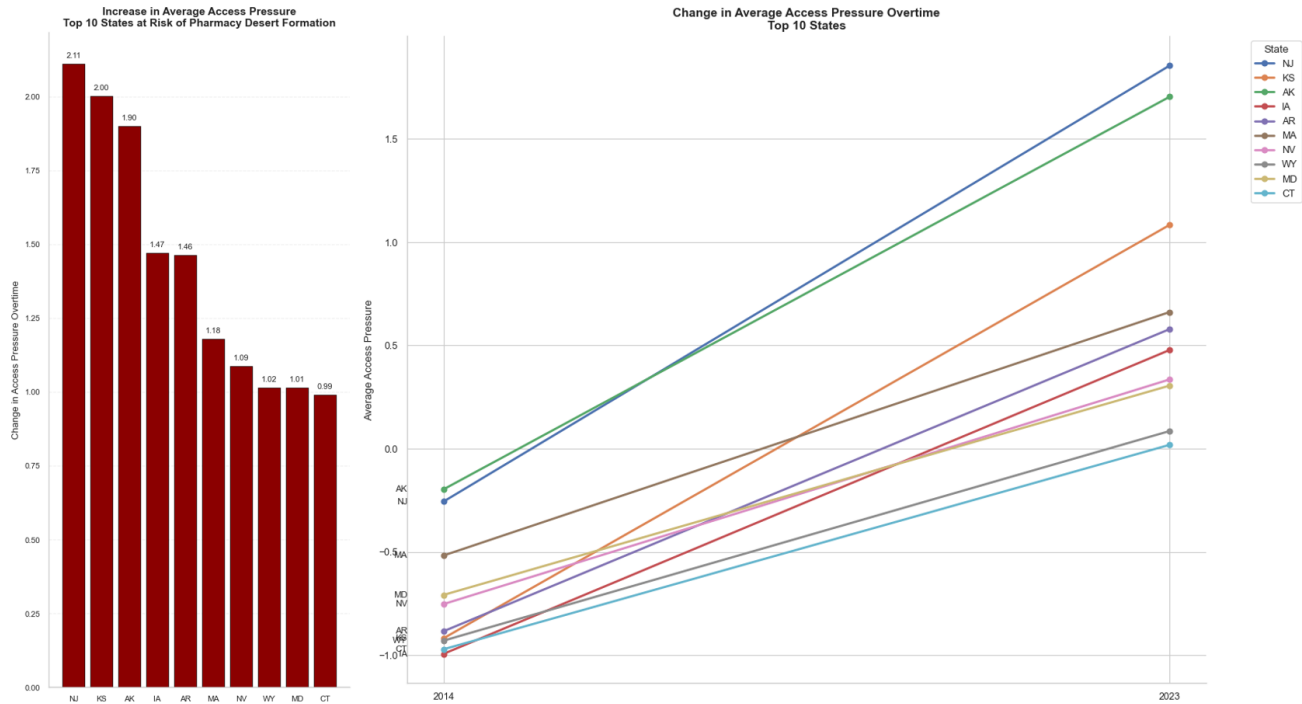


Fig. 3.

In Figure 4 below, the bar chart on the top left shows the counties of Lowndes (Georgia), Chautauqua (New York) and Newport News City (Virginia) being the counties with the highest increase in pharmacy access pressure over the study period. For the bar chart on the top right, the highest pharmacy access pressure change was recorded in 2019 for Chenango (New York), followed by Washington (Mississippi) in 2018 and Saline, IL (Illinois) in 2019. Comparing low risk versus high risk exposure to pharmacy desert formation in the bottoms bar chart, DC had 100% exposure to high risk (given that it has no counties) while Delaware had no counties exposed to high risk over the study period. The states of Alaska, New Vermont and Washington had next highest percentage of counties exposed to high risk.

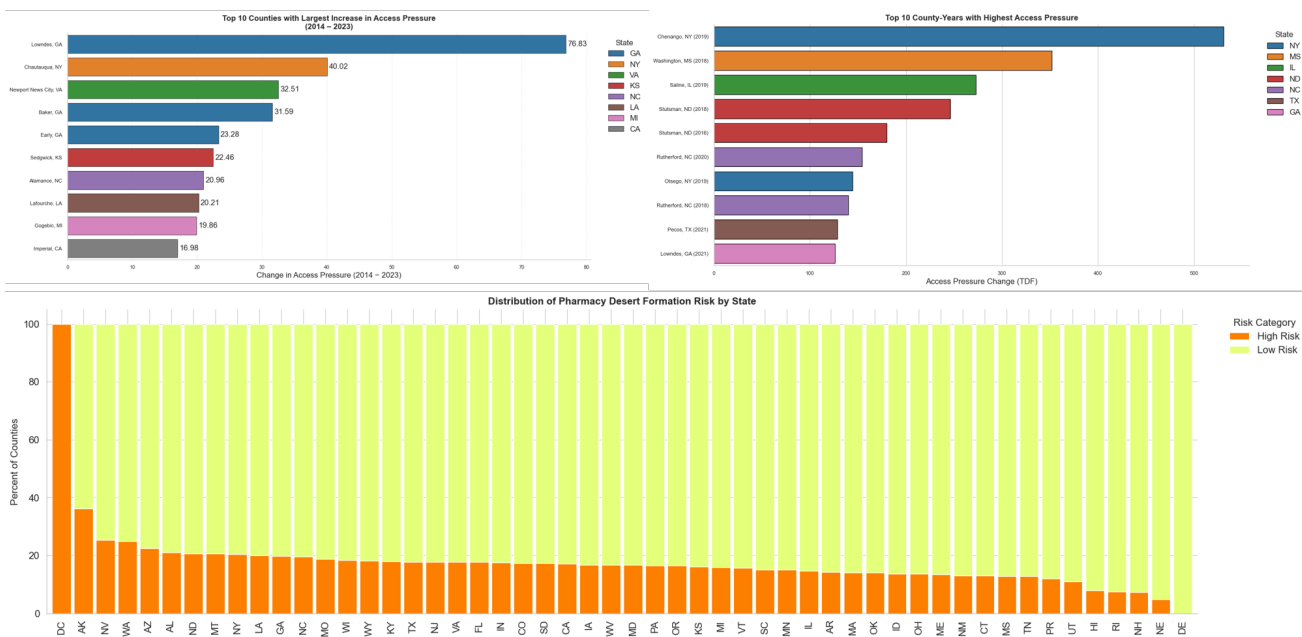


Fig. 4.

The correlation heatmap on the left of Figure 5 below shows that total claims and pharmacies per county year had very strong positive correlation at 0.93. Also, at 0.85, Health Factor Rank and Health Outcome Rank had a strong positive correlation. The trend analysis plot on the top left showed that the average number of beneficiaries per pharmacy had a rising trend overtime, from 2014 through to 2023. The trend analysis plot on the bottom right showed that the average unemployment rate had the same trend overtime for counties that had both high and low risk of pharmacy desert formation. Notably, the counties with high risk of pharmacy desert formation had on average higher unemployment rate, across all the years except 2018.

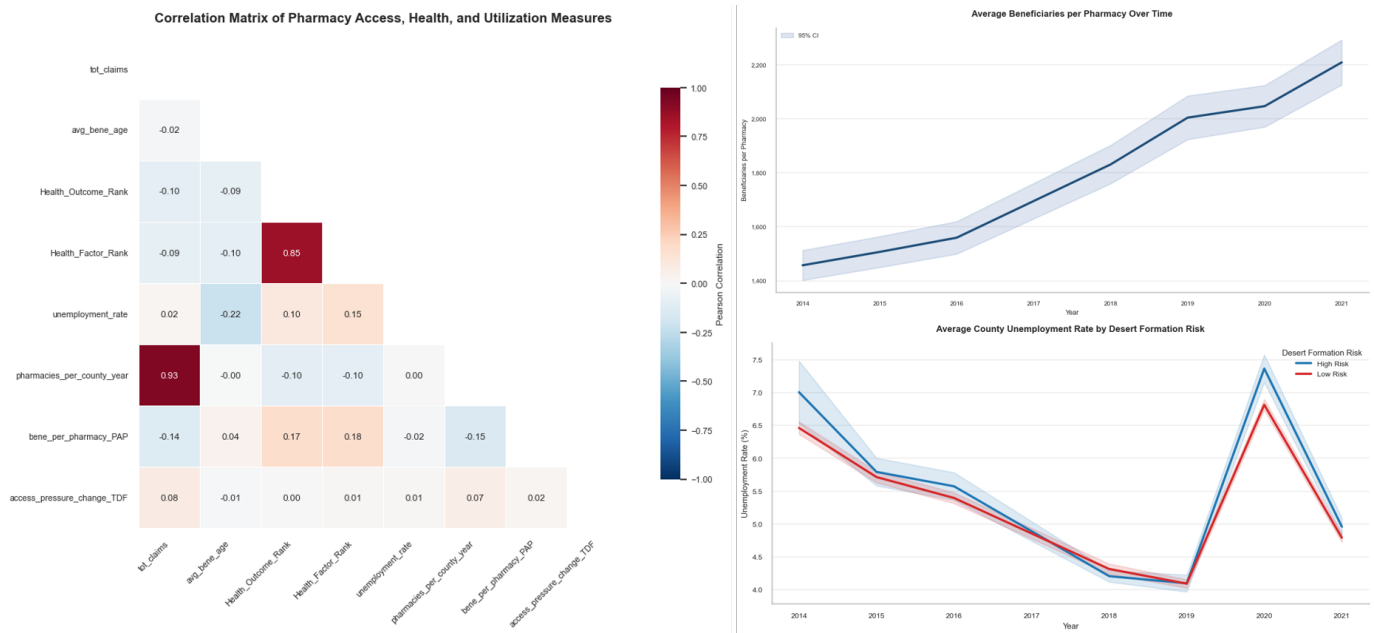


Fig. 5.

D. Temporal Modeling

The target for the modeling was Desert Formation Risk, risk of a county being a pharmacy desert across successive years; High Risk and Low Risk. The features were 'tot claims' (consumer demand for total number of customers), 'tot 30day fills' (service demand and profitability indicator), 'tot drug cost' (profitability indicator for cost of drugs bought), 'tot days supply' (operations indicator on ability to meet demand for drugs), 'avg bene age' (customer characteristic in terms of age), 'Health Outcome Rank' (the rank of how healthy the county in which a pharmacy is located is), 'Health Factor Rank' (the rank of the health risks for the county in which a pharmacy is located is), 'unemployment rate' (macroeconomic indicator for average purchasing power of population in the county) The XGBoost parameters were `XGBClassifier()` with `objective='binary:logistic'` and `eval metric='logloss'` while the LSTM parameters were `sequence=2`, `learning rate=0.001`, `epochs=100`, `criterion = nn.BCEWithLogitsLoss()`.

Table 1 below provides the model performance results for XGBoost and LSTM models. The results showed that XGBoost and LSTM correctly classified 82% and 83% of the exposure risk to pharmacy desert formation, respectively.

Model	Accuracy	Precision	Recall
XGBoost	0.82	0.75	0.82

Model	Accuracy	Precision	Recall
LSTM	0.83	0.64	0.83

TABLE I
XGBOOST AND LSTM MODEL PERFORMANCE

In Figure 6 below, the feature importance plots for the temporal models are presented. We note that for the XGBoost model in the bar chart on the left, tot drug cost, tot claims and tot days supply were the top three most important features. On the other hand, for the LSTM model in the bar chart on the right, tot drug cost, tot 30day fills and tot claims were the top three most important features.

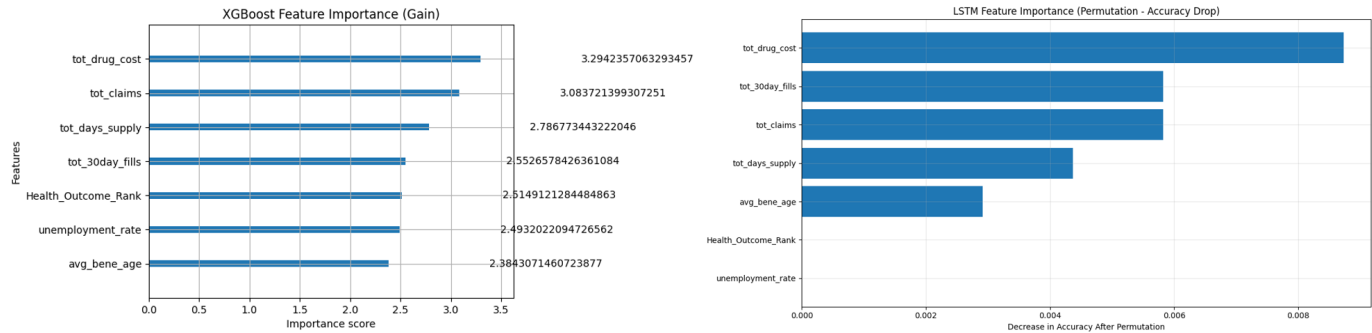


Fig. 6.

E. Spatial-Temporal Modeling: Spatial-Temporal Graph Neural Network (ST-GNN)

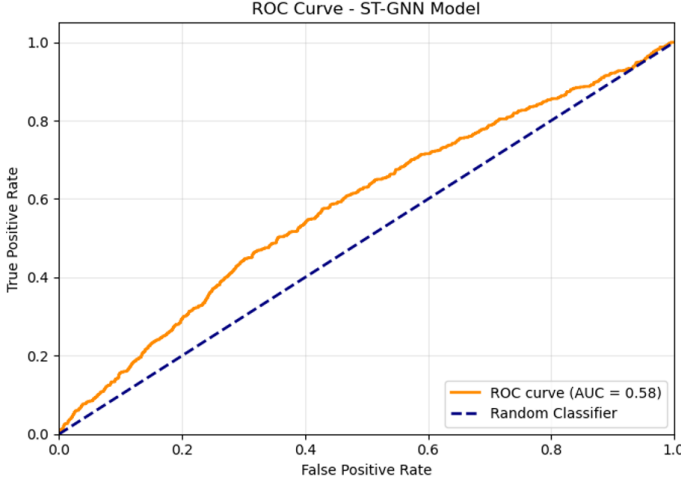
The summary of the ST-GNN model architecture is presented below in Table 2 with the model having 3 temporal blocks (GCN1, GCN2 and GCN3) and 1 Sequential Pool block with AdaptiveAvgPool1d layer.

Temporal Blocks			Sequential Pool
GCN1 (gc_conv): Linear(in_features=12, out_features=32, bias=True) (temporal_conv): Conv1d(32, 32, kernel_size=(3,), stride=(1,), padding=(1,)) (relu): ReLU() (dropout): Dropout(p=0.1, inplace=False) (attention): Linear(in_features=32, out_features=1, bias=True) (downsample): Linear(in_features=12, out_features=32, bias=True)	GCN2 (gc_conv): Linear(in_features=32, out_features=64, bias=True) (temporal_conv): Conv1d(64, 64, kernel_size=(3,), stride=(1,), padding=(1,)) (relu): ReLU() (dropout): Dropout(p=0.1, inplace=False) (attention): Linear(in_features=64, out_features=1, bias=True) (downsample): Linear(in_features=32, out_features=64, bias=True)	GCN3 (gc_conv): Linear(in_features=64, out_features=64, bias=True) (temporal_conv): Conv1d(64, 64, kernel_size=(3,), stride=(1,), padding=(1,)) (relu): ReLU() (dropout): Dropout(p=0.1, inplace=False) (attention): Linear(in_features=64, out_features=1, bias=True)	AdaptiveAvgPool1d(output_size=1) – Sequential - Linear(in_features=64, out_features=128, bias=True) - ReLU() - Dropout(p=0.3, inplace=False) - Linear(in_features=128, out_features=64, bias=True) - ReLU() - Linear(in_features=64, out_features=1, bias=True)

TABLE II
ST-GNN MODEL ARCHITECTURE

The target for the model was Desert Formation Risk, risk of a county being a pharmacy desert across successive years; High Risk and Low Risk. The ST-GNN parameters were sequence=3, learning rate=0.001, epochs=50, criterion = nn.BCEWithLogitsLoss(). The baseline features were 'tot claims' (consumer demand for total number of customers), 'tot 30day fills' (service demand and profitability indicator), 'tot drug cost' (profitability indicator for cost of drugs bought), 'avg bene age' (customer characteristic in terms of age), 'Health Outcome Rank' (the rank of how healthy the county in which a pharmacy is located is), 'Health Factor Rank' (the rank of the health risks for the county in which a pharmacy is located is), 'unemployment rate' (macroeconomic indicator for average purchasing power of population in the county). The spatial-temporal features were 'queen lag tot claims', 'queen lag tot drug cost', 'queen lag unemployment rate' - created using the Queens Contiguity weights; which are weights created based on how adjacent counties are to each other.

Figure 7 below shows the model performance summary for the ST-GNN model. The results in the table showed that the ST-GNN correctly classified 72% of the exposure risk to pharmacy desert formation. The ROC Curve, showed that the model had AUC = 0.58, implying a low discriminatory power for the model. The feature importance bar chart showed that queen lag unemployment rate, avg beneficiary age and tot 30day fills were the top three most important features.



Model	Accuracy	Precision	Recall
ST-GNN	0.72	0.26	0.25

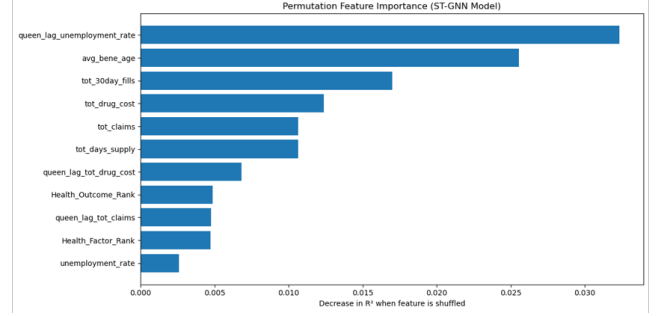


Fig. 7. ST-GNN Model Performance Summary

F. Higher Order Network Modeling: Hypergraph Neural Network (HGNN)

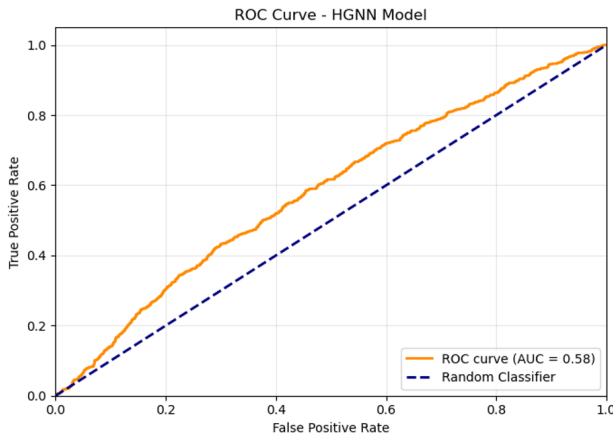
The summary of the HGNN model architecture is presented below in Table 3 with the model having 3 temporal blocks (HGCN1, HGCN2 and HGCN3) and 1 Sequential Pool block with AdaptiveAvgPool1d layer.

Temporal Blocks			Sequential Pool
HGCN1 (hg_linear1): Linear(in_features=9, out_features=32, bias=True) (hg_linear2): Linear(in_features=9, out_features=32, bias=True) (temporal_conv): Conv1d(32, 32, kernel_size=(3,), stride=(1,), padding=(1,)) (relu): ReLU() (dropout): Dropout(p=0.1, inplace=False) (attention): MultiheadAttention((out_proj): NonDynamicallyQuantizableLinear(in_features=32, out_features=32, bias=True)) (downsample): Linear(in_features=9, out_features=32, bias=True)	HGCN2 (hg_linear1): Linear(in_features=32, out_features=64, bias=True) (hg_linear2): Linear(in_features=32, out_features=64, bias=True) (temporal_conv): Conv1d(64, 64, kernel_size=(3,), stride=(1,), padding=(1,)) (relu): ReLU() (dropout): Dropout(p=0.1, inplace=False) (attention): MultiheadAttention((out_proj): NonDynamicallyQuantizableLinear(in_features=64, out_features=64, bias=True)) (downsample): Linear(in_features=32, out_features=64, bias=True)	HGCN3 (hg_linear1): Linear(in_features=64, out_features=64, bias=True) (hg_linear2): Linear(in_features=64, out_features=64, bias=True) (temporal_conv): Conv1d(64, 64, kernel_size=(3,), stride=(1,), padding=(1,)) (relu): ReLU() (dropout): Dropout(p=0.1, inplace=False) (attention): MultiheadAttention((out_proj): NonDynamicallyQuantizableLinear(in_features=64, out_features=64, bias=True))	AdaptiveAvgPool1d(output_size=1) – Sequential - Linear(in_features=64, out_features=128, bias=True) - ReLU() - Dropout(p=0.3, inplace=False) - Linear(in_features=128, out_features=64, bias=True) - ReLU() - Linear(in_features=64, out_features=1, bias=True)

TABLE III
HGNN MODEL ARCHITECTURE

The target for the model was Desert Formation Risk, risk of a county being a pharmacy desert across successive years; High Risk and Low Risk. The HGNN parameters were sequence=2, learning rate=0.001, epochs=50, criterion = nn.BCEWithLogitsLoss(). The baseline features were 'tot claims' (consumer demand for total number of customers), 'tot 30day fills' (service demand and profitability indicator), 'tot drug cost' (profitability indicator for cost of drugs bought), 'avg bene age' (customer characteristic in terms of age), 'Health Outcome Rank' (the rank of how healthy the county in which a pharmacy is located is), 'Health Factor Rank' (the rank of the health risks for the county in which a pharmacy is located is), 'unemployment rate' (macroeconomic indicator for average purchasing power of population in the county).

Figure 8 below shows the model performance summary for the HGNN model. The results in the table showed that the HGNN correctly classified 72% of the exposure risk to pharmacy desert formation. The ROC Curve, showed that the model had AUC = 0.58, implying a low discriminatory power for the model. The feature importance bar chart showed that unemployment rate, Health Factor Rank and avg beneficiary age were the top three most important features.



Model	Accuracy	Precision	Recall
HGNN	0.72	0.23	0.26

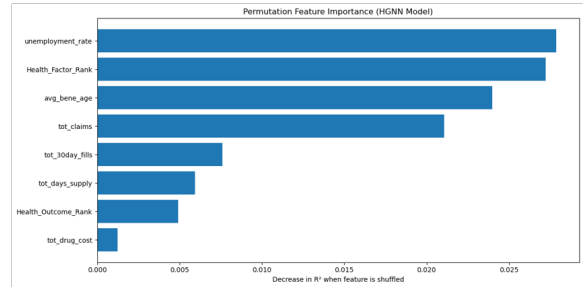
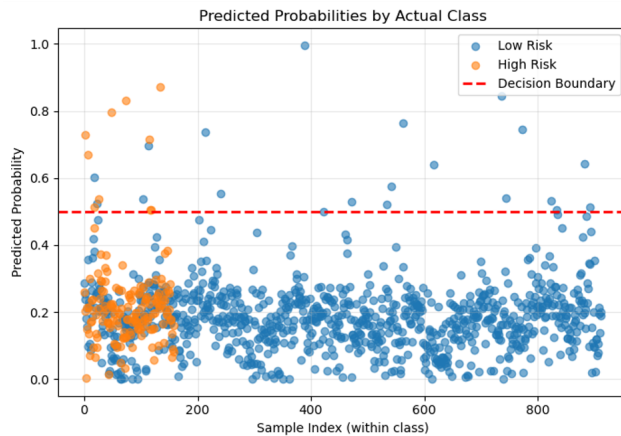


Fig. 8. HGNN Model Performance Summary

G. Tensor Train Network (TT)

The model structure for the TT was such that it had a TTClassifier with a TTLayer(), Sequential in nature containing ReLU(), Linear(in features=32, out features=16, bias=True), ReLU(), Linear(in features=16, out features=1, bias=True) and Sigmoid(). The target for the TT was Desert Formation Risk, risk of a county being a pharmacy desert across successive years; High Risk and Low Risk. The TT parameters were learning rate=0.001, epochs=200, criterion = nn.BCELoss(). The baseline features were 'tot claims' (consumer demand for total number of customers), 'tot 30day fills' (service demand and profitability indicator), 'tot drug cost' (profitability indicator for cost of drugs bought), 'avg bene age' (customer characteristic in terms of age), 'Health Outcome Rank' (the rank of how healthy the county in which a pharmacy is located is), 'Health Factor Rank' (the rank of the health risks for the county in which a pharmacy is located is), 'unemployment rate' (macroeconomic indicator for average purchasing power of population in the county). The spatial-temporal features were 'queen lag tot claims', 'queen lag tot drug cost', 'queen lag unemployment rate' - spatial features created using the Queens Contiguity weights which are weights created based on how adjacent counties are to each other. Figure 9 below shows the model performance summary for the TT model. The results in the table showed that the TT correctly classified 84% of the exposure risk to pharmacy desert formation. The predicted probabilities by Actual Cases plot show high false negative rate for the model. The feature importance bar chart showed that tot days supply, tot drug cost, tot 30day fills were the top three most important features.



Model	Accuracy	Precision	Recall
TT	0.84	0.35	0.06

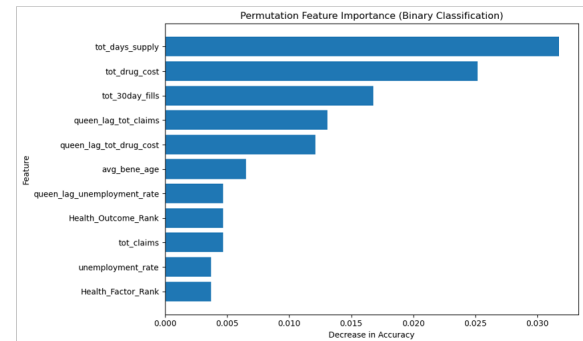


Fig. 9. TT Model Performance Summary

The table and bar chart in Figure 10 below show that among the advanced models, only the TT performs slightly better in terms of accuracy, as compared to the baselines.

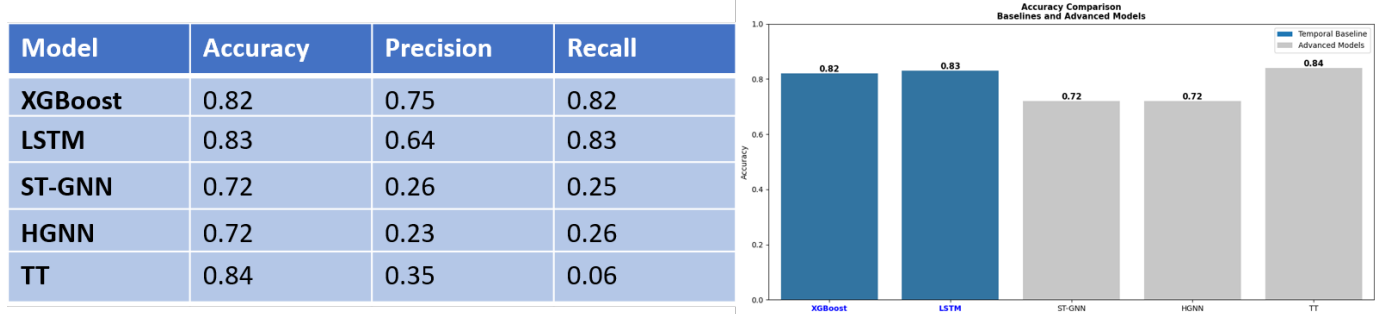


Fig. 10. Model Performance Comparison: Baseline and Advanced Models

VI. FINDINGS AND CONCLUSIONS

The states of New Jersey, Kansas and Alaska lead in highest increase in pharmacy access pressure over the study period; with the counties of Lowndes (Georgia), Chautauqua (New York) and Newport News City (Virginia) being the counties with the highest increase in pharmacy access pressure over the study period. The highest pharmacy access pressure change was recorded in 2019 for Chenago (New York), followed by Washington (Mississippi) in 2018 and Saline (Illinois) in 2019. Comparing low risk versus high risk exposure to pharmacy desert formation. DC had 100% exposure to high risk (given that it has no counties) while Delaware had no counties exposed to high risk over the study period. The states of Alaska, New Vermont and Washington had next highest percentage of counties exposed to high risk.

As baselines, the Temporal models had good predictive performance with XGBoost and LSTM correctly classifying 82% and 83% respectively for the exposure risk to pharmacy desert formation. The inclusion of spatial features and graph modeling in the ST-GNN did not improve on the exposure risk classification with a drop to 72% in accuracy as compared to the 2 baseline models. The hyper graph modelling in the HGNN also did not improve on the exposure risk classification with a drop to 72% in accuracy as compared to the 2 baseline models. The Tensor Train Network (TT) however presented an improvement on the 2 baseline models with 84% correct classification of exposure risk to pharmacy desert formation. However, accuracy alone was not sufficient because the Tensor Train model showed weak High-Risk recall. Therefore, XGBoost remained the best practical baseline because it provided balanced performance and interpretable feature importance.

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